DEVELOPMENT OF AN

Ethernet-Based Vehicle Monitoring System

Using Linux Socket Programming in C

Test Procedures, Observations and Evidence

Project platform: Pop!_OS / Linux

Communication: TCP and UDP sockets

1. Purpose

This report defines the validation scenarios for the Vehicle Monitoring System. Actual outputs, packet captures, and screenshots should be retained as evidence. A scenario must not be marked as passed until its result has been observed and recorded.

2. Test Environment

Item	Configuration
Operating system	Pop!_OS 24.04 LTS
Language	C
Build tools	GCC and Make
Capture tool	Wireshark / dumpcap
UDP endpoint	127.0.0.1:5005
TCP endpoint	127.0.0.1:5006

3. Validation Scenarios

Scenario	Procedure	Evidence to record
1. Normal communication	Start dashboard and vehicle in the same protocol mode.	Sender and receiver output; packet capture.
2. Dashboard unavailable	Stop dashboard and attempt UDP sending and TCP connection.	Actual sender output and TCP connection error.
3. Extended transmission	Run both applications for a defined period, e.g. 60 seconds.	Duration, messages sent, messages received, observed issues.
4. Invalid configuration	Provide an invalid IPv4 address and invalid port.	Actual exit status and error messages.
5. TCP/UDP comparison	Capture both modes and compare packet behaviour.	Handshake, data, teardown, ports and payload evidence.

4. Results Record

Complete the following table using the actual test output. Replace each pending entry with Pass or Fail and reference the associated evidence.

Test	Status	Observed result / evidence
Normal UDP	Pending	Record receiver output and UDP capture.
Normal TCP	Pending	Record receiver output and TCP capture.
Dashboard unavailable	Pending	Record UDP and TCP behaviour.
Extended duration	Pending	Record duration and actual message counts.
Invalid IP	Pending	Record actual program output.

Test	Status	Observed result / evidence
Invalid port	Pending	Record actual program output.
Wireshark analysis	Pending	Reference saved captures and screenshots.

5. Packet Analysis Checklist

- UDP: source IP, destination IP, source port, destination port, payload.
- TCP: SYN, SYN-ACK, ACK, data transfer, and connection termination.
- Check that captured packets correspond to the application ports.
- Save genuine screenshots and packet captures in the repository.

6. Limitations

Loopback testing validates local socket communication, not physical Ethernet hardware, external network performance, or communication between separate ECUs. Test outcomes depend on the actual execution environment.

7. Conclusion

Finalize this report after all tests have been run and evidence has been reviewed. Pending scenarios must remain clearly identified until verified.